

CO4_AT2 – INDUSTRY PROBLEM SOLVING TASK: ANSWERS

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1. Autonomous Vehicle Vision System Sélection

Constraints

- Accuracy must be $\geq 90\%$ (below this, the system fails).
- Latency must be ≤ 70 ms (above this, it is unacceptable).
- Computational load must fit moderate available resources.

Data

Method	Accuracy	Latency	Computation
Viola-Jones	82%	30 ms	Low
YOLO	93%	60 ms	Moderate
Deep Learning	97%	90 ms	High

Constraint-by-Constraint Evaluation

- Viola-Jones: Accuracy = 82%, which is below the required 90% threshold → FAILS the accuracy constraint, even though its latency (30 ms) and computation (low) are excellent. It is eliminated immediately.
- YOLO: Accuracy = 93% ($\geq 90\%$, passes). Latency = 60 ms (≤ 70 ms, passes). Computation = moderate, which matches the vehicle's available resources exactly → PASSES all three constraints.
- Deep Learning: Accuracy = 97% ($\geq 90\%$, passes). Latency = 90 ms, which exceeds the 70 ms limit → FAILS the latency constraint. It also requires high computation, exceeding the moderate resource limit → FAILS the computation constraint as well.

Comparison

Viola-Jones offers speed and efficiency but cannot meet the minimum accuracy needed for safe autonomous driving. Deep Learning offers the highest accuracy but is too slow and too resource-intensive for the vehicle's real-time, moderate-hardware requirements. YOLO is the only method that simultaneously satisfies the accuracy floor, the latency ceiling, and the available computational budget.

Decision and Justification

YOLO should be selected for the autonomous vehicle vision system.

- Logical reasoning: An autonomous vehicle needs both safety (high accuracy) and real-time responsiveness (low latency) simultaneously — a method must clear both bars, not just one. Viola-Jones clears the latency bar but not the accuracy bar; Deep Learning clears the accuracy bar but not the latency or computation bars.
- YOLO is the only option that satisfies all three constraints at once: 93% accuracy exceeds the 90% safety threshold, 60 ms latency stays under the 70 ms real-time limit, and its moderate computational need matches the vehicle's available hardware.
- Practical implication: Choosing Viola-Jones risks missed or incorrect object detections (pedestrians, vehicles, obstacles) at highway speeds, which is unacceptable for safety. Choosing Deep Learning risks delayed reactions (90 ms latency) that could be critical in collision-avoidance scenarios, and it would also require hardware upgrades the vehicle does not support.

Conclusion: YOLO provides the best-engineered trade-off between accuracy, speed, and hardware compatibility, making it the only feasible and safe choice under the given constraints.

2. Smart Factory Product Inspection Selection

Constraints

- Accuracy must be $\geq 95\%$ (below this, defects may be missed).
- Processing time must be ≤ 50 ms (above this, production is affected).
- Hardware supports moderate computation.

Data

Method	Accuracy	Processing Time	Computation
Traditional Image Analysis	91%	25 ms	Low
YOLO	96%	45 ms	Moderate
Deep Learning	98%	75 ms	High

Constraint-by-Constraint Evaluation

- Traditional Image Analysis: Accuracy = 91%, below the required 95% $\rightarrow$ FAILS the accuracy constraint, despite very fast processing (25 ms) and low computation. Missed defects on a production line make this unsuitable regardless of its speed advantage.
- YOLO: Accuracy = 96% ($\geq 95\%$, passes). Processing time = 45 ms (≤ 50 ms, passes). Computation = moderate, matching the available hardware $\rightarrow$ PASSES all three constraints.
- Deep Learning: Accuracy = 98% ($\geq 95\%$, passes). Processing time = 75 ms, exceeding the 50 ms limit $\rightarrow$ FAILS the processing-time constraint. It also demands high computation, beyond the moderate hardware available $\rightarrow$ FAILS the computation constraint too.

Comparison

Traditional Image Analysis is the fastest and lightest but is not accurate enough to reliably catch defective products, which directly undermines quality control. Deep Learning is the most accurate but is too slow for the production line's throughput requirement and needs hardware beyond what is available. YOLO is the only method that meets the accuracy floor, stays within the processing-time limit, and fits the moderate computational budget.

Decision and Justification

YOLO should be selected for the smart factory product inspection system.

- Logical reasoning: A production-line inspection system must reliably catch defects (accuracy) without slowing down the line (processing time), and it must run on the hardware actually installed (computation). Traditional Image Analysis satisfies the speed and hardware constraints but not accuracy; Deep Learning satisfies accuracy but not speed or hardware.
- YOLO is the only method that clears all three constraints simultaneously: 96% accuracy exceeds the 95% quality threshold, 45 ms processing time stays under the 50 ms limit, and its moderate computational requirement matches the available hardware.

- Practical implication: Using Traditional Image Analysis risks defective products passing inspection undetected, leading to quality failures and potential recalls. Using Deep Learning would slow the production line (75 ms per item) and require costly hardware upgrades to support high computation, increasing cost and reducing throughput.

Conclusion: YOLO offers the best balance of accuracy, speed, and hardware compatibility for real-time defect detection, making it the only method that satisfies all deployment constraints for the smart factory.